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Lab Task 7

①

FCFS (first come first serve):

P ₁	P ₂	P ₃	P ₄	
0	6	14	21	24

Process	AT	CT	BT
P ₁	0	6	6
P ₂	2	14	8
P ₃	4	21	7
P ₄	6	24	3

① P₁:

Completion time = 6

$$TAT = \text{Completion time} - \text{Arrival time}$$

$$WT = TAT - BT$$

$$TAT = 6 - 0 = 6$$

$$WT = 6 - 6 = 0$$

② P₂:

$$CT = 14$$

$$TAT = 14 - 2 = 12$$

$$WT = 12 - 8 = 4$$

$$P_3: CT = 21$$

$$TAT = 21 - 4 = 17$$

$$WT = 17 - 7 = 10$$

$P_4:$

$$CT = 24$$

$$TAT = 24 - 6 = 18$$

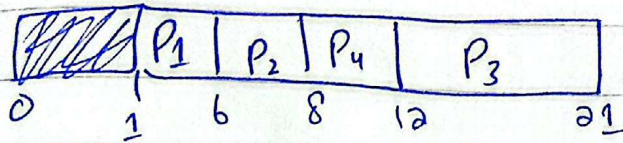
$$WT = 18 - 3 = 15$$

$$\text{Average } WT = \frac{0 + 4 + 10 + 5}{4} = 7.25$$

$$\text{Average } TAT = \frac{6 + 12 + 17 + 18}{4} = 13.25$$

CTC (check-out Tab Cost) / (Non-promotion)

Q) SBJF (shortest Job first):



~~But~~ Completion time = start + Burst time

TAT = Completion time - Arrival time

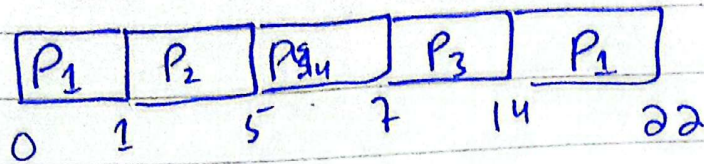
WT = TAT - BT

Process	AT	BT	CT	TAT	WT
P ₁	1	5	6	5	0
P ₂	3	2	8	5	3
P ₃	5	9	21	16	7
P ₄	6	4	12	6	2

$$\text{Average TAT} = \frac{5 + 5 + 16 + 6}{4} = 8$$

$$\text{Average WT} = \frac{0 + 3 + 7 + 2}{4} = 3$$

(3) SRTF (Preemptive),



$$\begin{aligned} \overline{TAT} &= CT - AT \\ WT &= \overline{TAT} - BT \end{aligned}$$

①

Process	AT	BT	CT	TAT	WT
P ₁	0	9	22	22	13
P ₂	1	4	5	4	0
P ₃	2	7	14	12	5
P ₄	3	2	7	4	2

$$\text{Average } \overline{TAT} = \frac{22 + 4 + 12 + 4}{4} = 10.5$$

$$\text{Average } WT = \frac{13 + 0 + 5 + 2}{4} = 5$$

②

④ Round Robin (RR) Scheduling

Time Quantum = 3 ms

$P_2 \quad P_3 \quad P_4 \quad P_1$

P_1 P_2 P_3 P_4 P_1 P_2 P_3 P_1 P_3 P_1
 0 3 6 9 12 15 17 20 23 25 26

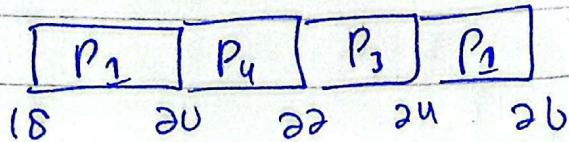
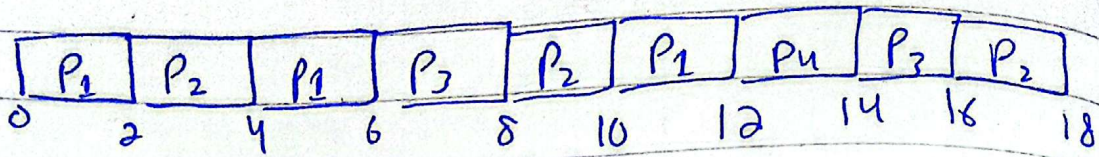
Process	AT	BT	CT	TAT	WT
P_1	0	10	26	26	16
P_2	1	5	17	16	11
P_3	2	8	25	23	15
P_4	3	3	10	9	6

$$\text{Average TAT} = \frac{26 + 16 + 23 + 9}{4} = 18.5$$

$$\text{Average WT} = \frac{16 + 11 + 15 + 6}{4} = 12$$

⑤ Round Robin (RR):

Time Quantum = 2 ms



$$TAT = CT - AT$$

$$WT = TAT - BT$$

①

Process	AT	BT	TAT	CT	WT
P1	0	12	30	30	18
P2	2	6	16	18	10
P3	4	8	24	28	16
P4	6	4	16	22	12

$$\text{Average TAT} = \frac{30 + 16 + 24 + 6}{4} = 21.5$$

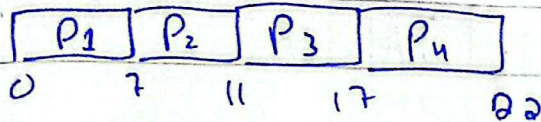
$$\text{Average WT} = \frac{18 + 10 + 16 + 12}{4} = 14$$

Task 8:

Priority scheduling (Non-preemptive)

As P_1 arrive at $t=0$

So,

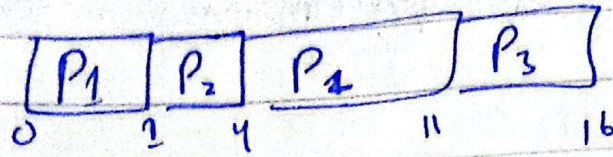


Process	AT	BT	CT	TAT	WT	Priority
P_1	0	7	7	7	0	3
P_2	1	4	11	10	6	1
P_3	2	6	17	15	9	2
P_4	3	5	22	19	14	4

$$\text{Average Turnaround Time} = \frac{7 + 10 + 15 + 19}{4} = 12.75$$

$$\text{Average waiting time} = \frac{0 + 6 + 9 + 14}{4} = 7.25$$

Task 7: Priority scheduling (Preemptive)



Process	AT	BT	Priority	CT	TAT	WT
P ₁	0	8	2	11	11	3
P ₂	1	3	1	4	3	0
P ₃	2	5	3	16	14	9
P₄						

①

$$\text{Average TAT} = \frac{11 + 3 + 14}{3} = 9.33$$

$$\text{Average WT} = \frac{3 + 0 + 9}{3} = 4.00$$

②